

An Architecture for YANG-Push to Message Broker **Integration**

draft-ietf-nmop-yang-message-broker-integration-13

Motivation and architecture of a native
YANG-Push notifications and YANG Schema integration
into Message Broker and YANG Schema Registry

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An Architecture for YANG-Push to Message Broker Integration

Operational and Considerations Sections in -12

7. Security Considerations

In line with [Section 4.1.3](#) of [\[RFC3552\]](#), the NETCONF SSH transport session between YANG-Push Receiver and Publisher SHOULD be authenticated with publickey as defined in [Section 7](#) of [\[RFC4252\]](#). In order to discover the YANG-Push Subscription capabilities, the YANG module dependencies, the YANG modules and the Subscription state, read-only permission should be assigned to this user through centralized [Section 6](#) of Tacacs+ Authorization [\[RFC3552\]](#) or through NACM [\[RFC8341\]](#) locally. Tacacs+ authorization MUST be secured with TLS 1.3 as described in [\[RFC8341\]](#). Depending on NACM permissions, only authorized YANG nodes are discoverable and subscribable by NETCONF and RESTCONF clients.

In line with [Section 4.5.2](#) of [\[RFC3552\]](#), the YANG-Push transport session SHOULD be secured with [DTLS 1.3](#) [\[RFC9147\]](#), [TLS 1.3](#) [\[RFC8446\]](#) or [SSH](#) [\[RFC4252\]](#) depending on transport protocol and Subscription type. The HTTPS and TCP transport sessions among YANG Message Broker Producers, Consumers and YANG Schema Registry, MUST be secured with [TLS 1.3](#) [\[RFC8446\]](#). Keys depending on key length should be frequently rotated.

8. Operational Considerations

The Data Product should be monitored for YANG schema integrity, YANG notification end-to-end delay and loss from YANG-Push Subscription to YANG Data Consumer. The YANG instance data can be validated at the YANG-Push Receiver or YANG Message Broker Consumer or both. Notifications or messages not complying to YANG schema SHOULD be logged describing the YANG schema error with the instance data together for troubleshooting purposes to the data owner defined in the Stream Catalog. YANG notification delay and loss can be measured at the YANG-Push Receiver and the YANG Message Broker Consumer by taking the [\[I-D.ietf-netconf-notif-envelope\]](#) defined event-time, hostname and sequence-number and the [\[I-D.ietf-nmop-message-broker-telemetry-message\]](#) defined data-collection-manifest and collection-timestamp into context. The Service Level Objective SHOULD be defined by the Data Product owner in the Stream Catalog and be notified and reported when violated.

A YANG-Push Receiver or a YANG Message Broker Consumer MAY generate a YANG Library where YANG deviations are explicitly listed. Therefore the YANG library between YANG-Push Receiver and a YANG Message Broker Consumer MAY be identical or not.

An Architecture for YANG-Push to Message Broker Integration

Changes in -13 and Next Steps

Many thanks to Heng Cui for contributing the operational and security consideration sections, Deepya Mandadi on sharpening the YANG deviations aspects of the document and Reshad on reviewing normative downward references.

Status

- Detail on how YANG deviations are handled in YANG-Push Receiver YANG library, YANG schema registry and Message Broker YANG library.
- Example how the message key and schema id is encoded in Message Broker header meta data. How message keys are crafted as defined in [draft-ietf-nmop-yang-message-broker-message-key](#).
- <https://datatracker.ietf.org/doc/html/draft-ietf-netconf-udp-notif> is now a normative downward reference.

Next Steps

- No action items pending.

Confluent Schema Registry is pluggable. Currently Supports AVRO, JSON Schema and Protobuf. The YANG support is being developed at [Yak24] as part of this architecture. Enable to register, obtain and compare [YSR24] YANG Schemas. One YANG Schema with all its augmentations is being registered per YANG-Push subscription ID along with metadata describing which YANG features were enabled for the registered YANG schema and its YANG deviations. For each YANG Schema a locally significant YANG Schema ID is being issued as described in step (7) in the workflow diagram. YANG module imports are being listed as references, regardless if the YANG module is an augment or a deviation. YANG features are listed as feature metadata.

A YANG-Push Receiver or a YANG Message Broker Consumer MAY generate a YANG Library where YANG deviations are explicitly listed. Therefore the YANG library between YANG-Push Receiver and a YANG Message Broker Consumer MAY be identical or not.

4.6. YANG Message Broker Producer

The previously issued YANG Schema ID is added to the Message Broker header of the YANG push push-update message previously described in Section 4.3 before serialized to a Message Broker topic in step (8) of the workflow diagram.

Section 2 of [I-D.ietf-nmop-message-broker-telemetry-message] defines the envelope schema of the message facilitating the YANG-Push and different types of provenance metadata.

Figure 10 represent the Message Key and the Message Broker headers with the Schema ID and content type of the Message and Figure 11 the Message itself.

NOTE: The Key is a two-line byte string; the line break between the node hostname and the YANG path is a literal LF (U+000A).

Key:
router-nyc-01
1042
/ietf-interfaces:interfaces/interface[name='eth0']

Headers:
schema-id: 1
content-type: application/yang-data+json

Figure 10: Telemetry Message Broker Key and Header Example

```
{  
  "ietf-telemetry-message:message": {  
    "network-node-manifest": {  
      "name": "pe1",
```

An Architecture for YANG-Push to Message Broker Integration

Normative References, State and Reviewers

Working Group	Document Name	Document State	Reviews
NETCONF	draft-ietf-netconf-notif-envelope-05	Shepherd writeup concluded	OPSDIR Early review by Joe Clarke YANGDOCTORS Early review (of -02) by Jürgen Schönwälder
NETCONF	draft-ietf-netconf-yang-library-augmentedby-18	RFC editor queue	OPSDIR Early review (of -08 and -14) by Sheng Jiang, Xiao Min YANGDOCTORS Early review (of -07 and -13) by Andy Bierman SECDIR review (of -14 and -15) by Rich Salz
NETCONF	draft-ietf-netconf-yang-notifications-versioning-14	Submitted to IESG for Publication	YANGDOCTORS Early review (of -09) by Robert Wills OPSDIR review (of -10) by Gabriele Galimberti
NETCONF	draft-ietf-netconf-yp-transport-capabilities-06	Submitted to IESG for Publication	OPSDIR IETF Last Call review by Xiao Min TSVART IETF Last Call review by Tommy Pauly YANGDOCTORS IETF Last Call review by Jan Lindblad OPSDIR Early review (of -03) by Xiao Min YANGDOCTORS Early review (of -01) by Jan Lindblad SECDIR IETF Last Call Review due 2025-10-18
NMOP	draft-ietf-nmop-yang-message-broker-integration-13	WG Document (request for working group last call at IETF 124)	
NMOP	draft-ietf-nmop-message-broker-telemetry-message-04	WG Document (request for working group last call at IETF 125)	YANGDOCTORS Early review (of -02) by Martin Björklund

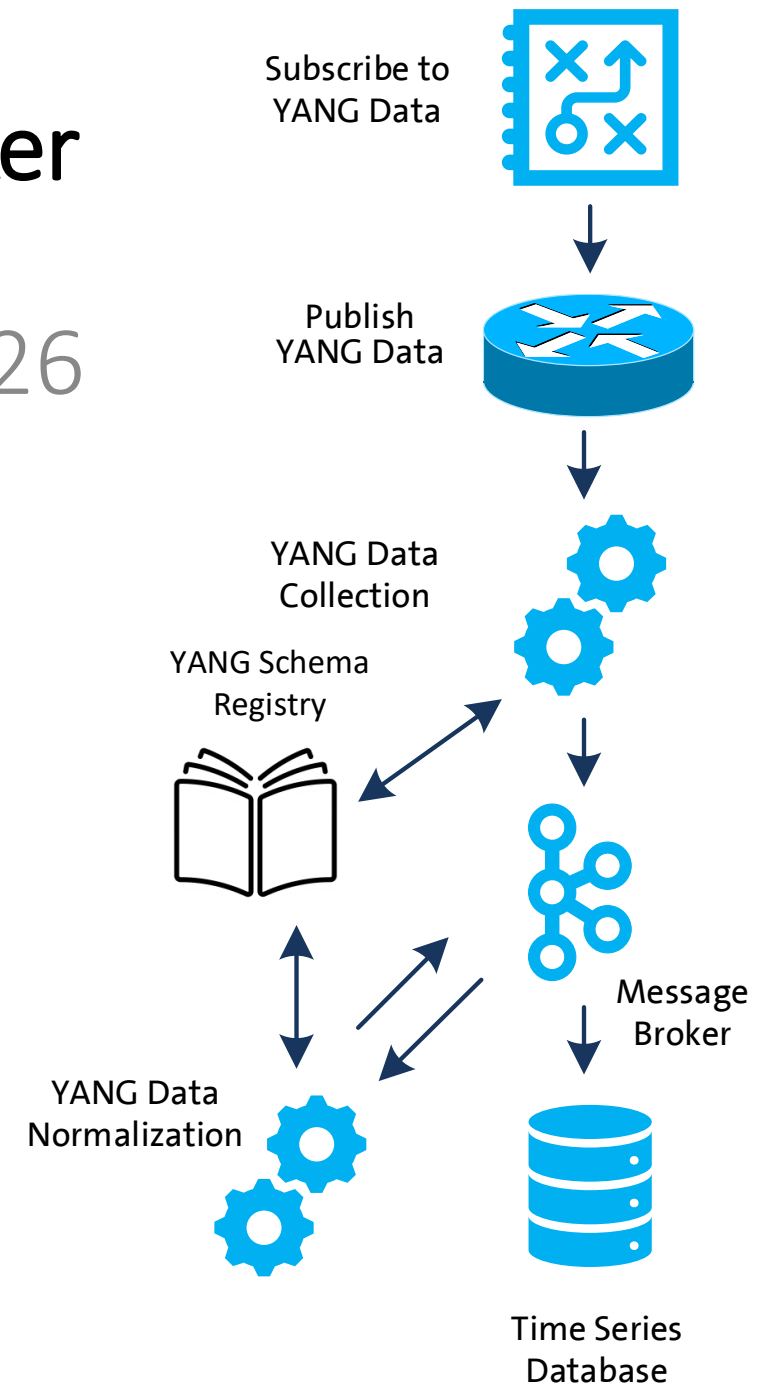
An Architecture for YANG-Push to Message Broker Integration

Informative References, State and Reviewers

Working Group	Document Name	Document State	Reviews
NETCONF	draft-ietf-netconf-udp-notif-25	WIT AD review received.	OPSDIR Early review (of -21) by Tina Tsou YANGDOCTORS Early review (of -20) by Jürgen Schönwälder TSVART Early review (of -11) by Michael Tüxen TSVART Early Review due 2025-10-23
NETCONF	RFC 9984	RFC	OPSDIR Early review (of -07) by Ran Chen TSVART Early review (of -06) by Joerg Ott YANGDOCTORS Early review (of -06) by Jürgen Schönwälder GENART review (of -09) by Elwin Davies DNSDIR review (of -09) by Peter Van Dijk
NETCONF	draft-ietf-netconf-distributed-notif-19	WIT AD review received.	OPSDIR Early review (of -14) by Jürgen Schönwälder YANGDOCTORS Early review (of -13 and -17) by Martin Björklund
NETMOD	draft-ietf-nmop-yang-anydata-validation-00	Reviews from Rob Wilton and James Cumings received.	

Validate YANG-Push to Message Broker End-To-End Data Processing Chain

IETF 126 Hackathon, July 18/19th 2026



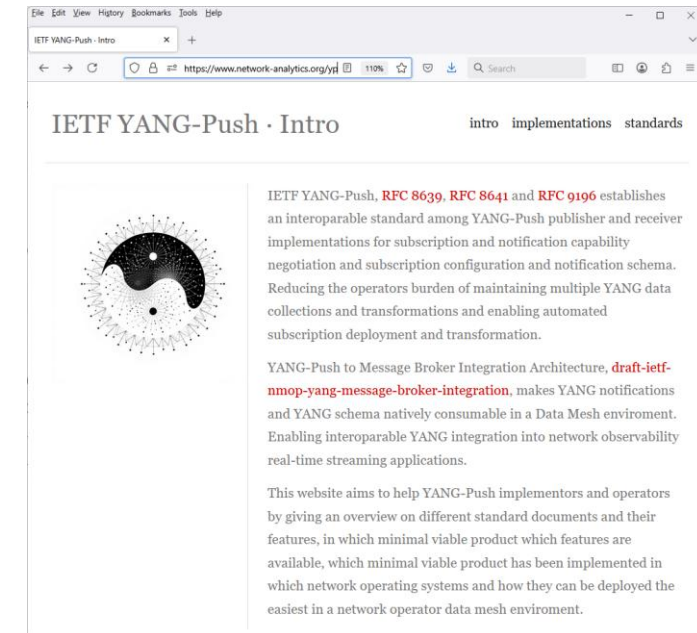
Hackathon Plan, Software and Website

Test Plan

Validate and verify

- 5 YANG-Push Publishers
- 2 YANG-Push Receivers
- 2 YANG-Push Network Telemetry Message
- 1 YANG Message Broker Producer and Schema Registry
- 3 YANG Message Broker Consumers

implementation in YANG data integration automation. Subscribe to YANG data on YANG-Publisher, obtain and register all YANG modules necessary to build YANG schema tree, register YANG schemas to Schema Registry and verify YANG notifications against scheme trees and produce and consume from Message Broker.



Software

<https://www.network-analytics.org/yp/how-to-deploy.html>

- YANG-Push Publisher - Cisco IOS XR, 6WIND VSR, Huawei NE (Router), MA (OLT), Arrcus Arcos and Ductus DMAP
- YANG-Push Receiver – [NetGauze™](#) and [Pmacct](#)
- YANG Message Broker Producer – [NetGauze™](#)
- YANG Message Broker Consumer – [NetGauze™](#), Ciena Blue Planet UAA Workflow Engine, Cisco Crosswork Ingestion Service
- YANG Schema Registry – [Apache Kafka YANG Plugin](#)
- Libyang - <https://github.com/CESNET/libyang/releases/tag/v5.4.9>
- Yangkit - <https://github.com/yang-central/yangkit/>
- Wireshark - [udp-notif dissector](#)

Hackathon – Repositories

The results of the end-to-end tests

- Netconf RPCs and YANG-Push JSON and CBOR encoded notifications with wire captures from receiver
- YANG-Push Receiver cache content generation with YANG library context, YANG modules, metadata and notification schema validation logs
- YANG Message Broker Producer serialization logs
- YANG Schema Registry schema examples
- YANG Message Broker Consumer, serialization logs, YANG library context, YANG modules and message schema validation

ietf-network-analytics-document-status / 126 / Hackathon / 

 graf3net ciena blueplanet uaa we b449565 · 1 minute ago  History

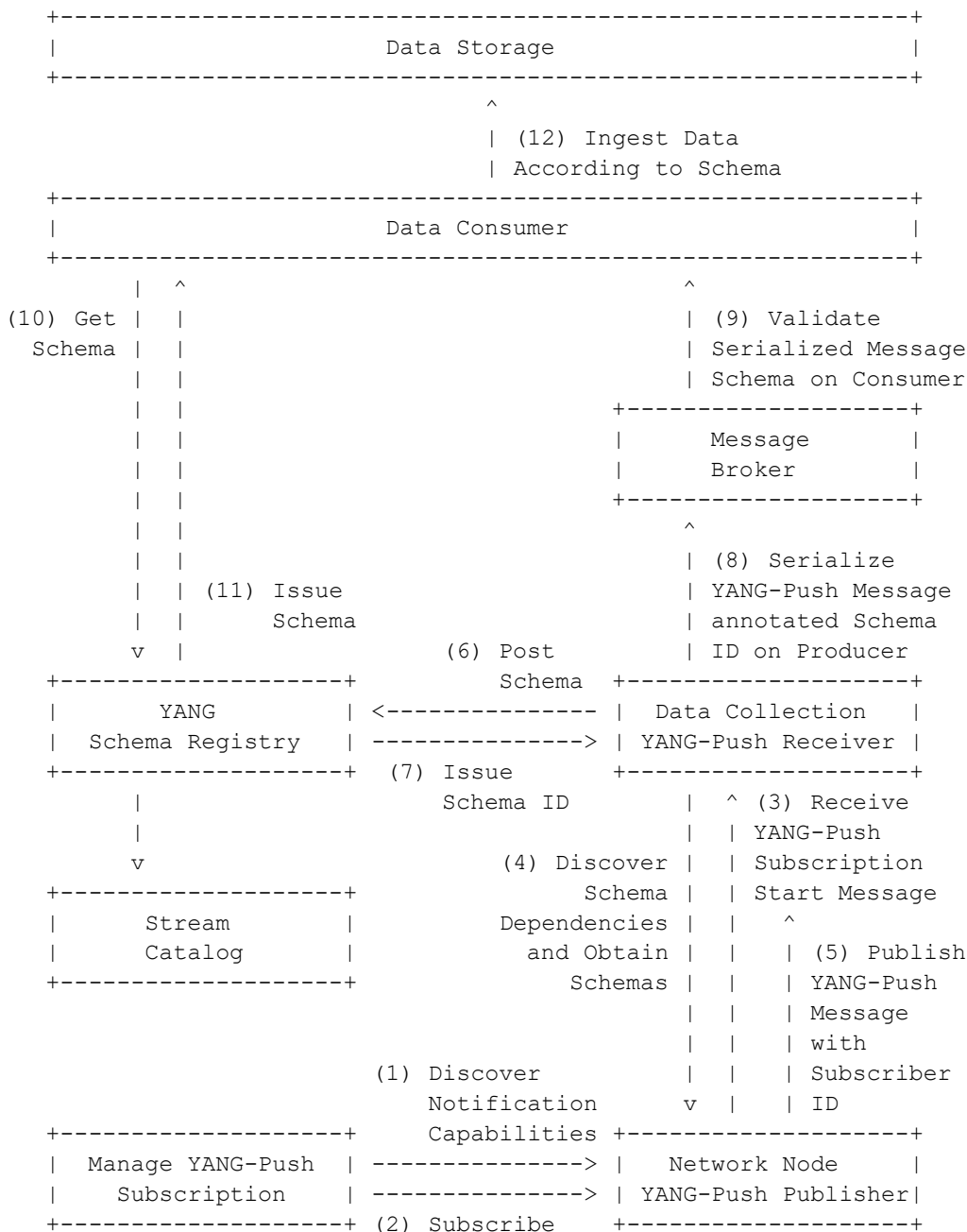
Name	Last commit message	Last commit date
 ..		
 mb-alive-test-result	ciena blueplanet uaa we	1 minute ago
 mb-test-cases	initial	last month
 mb-test-setup	nit	3 weeks ago
 yangpush-publisher	updated	7 minutes ago

<https://github.com/network-analytics/ietf-network-analytics-document-status/tree/main/126/Hackathon/>

An Architecture for YANG-Push to Message Broker Integration

draft-ietf-nmop-yang-message-broker-integration
 draft-ietf-nmop-message-broker-telemetry-message
 draft-ietf-nmop-yang-message-broker-message-key
 draft-netana-nmop-message-broker-bmp-telemetry-message

- Subscription to YANG Notifications
[RFC 8639](#)
- Subscription to YANG Notifications for Datastore Updates
[RFC 8641](#)
- UDP-based Transport for Configured Subscriptions
[draft-ietf-netconf-udp-notif](#)
- Subscription to Distributed Notifications
[draft-ietf-netconf-distributed-notif](#)
- Extensible YANG Model for YANG-Push Notifications
[draft-ietf-netconf-notif-envelope](#)
- Support of Versioning in YANG Notifications Subscription
[draft-ietf-netconf-yang-notifications-versioning](#)
- YANG Modules Describing Capabilities for Systems and Datastore Update Notifications
[RFC 9196](#)
- YANG Notification Transport Capabilities
[draft-ietf-netconf-yp-transport-capabilities](#)
- YANG Library
[RFC 8525](#)
- Augmented-by Addition into the IETF-YANG-Library
[draft-ietf-netconf-yang-library-augmentation](#)
- Validating anydata in YANG Library context
[draft-ietf-netmod-yang-anydata-validation](#)
- Encoding of Data Modeled with YANG in the CBOR
[RFC 9254](#)



YANG-Push Publisher - Subscription

- Netconf RPC calls to create, delete and obtain YANG-Push subscriptions and receiver configurations.

Name	Last commit message	Last commit date
..		
edit-config-yp-subscription-receiver-add.xml	testcases updated	last month
edit-config-yp-subscription-receiver-delete.xml	testcases updated	last month
get-yp-subscription.xml	testcases updated	last month
readme.md	testcases updated	last month

readme.md	✎ ☰
Netconf Interactive Login <pre>netconf-console2 --host=10.190.64.79 --port=830 --db=running -i --user=daisy1 --privKeyFile=/etc/daisy/netg</pre>	
Netconf Get YANG-Push subscriptions <pre>netconf-console2 --host=10.190.64.79 --port=830 --db=running --user=daisy1 --privKeyFile=/etc/daisy/netgau:</pre>	
Netconf Create YANG-Push subscriptions with receiver <pre>netconf-console2 --host=10.190.64.79 --port=830 --db=running --user=daisy1 --privKeyFile=/etc/daisy/netgau:</pre>	
Netconf Delete YANG-Push subscriptions with receiver <pre>netconf-console2 --host=10.190.64.79 --port=830 --db=running --user=daisy1 --privKeyFile=/etc/daisy/netgau:</pre>	

YANG-Push Publisher - Notifications

- Subscription state change notifications as JSON and PCAP.
- Push update and change-update notifications as JSON and PCAP.

Name	Last commit message	Last commit date
..		
1-push-update.json	testcases updated	last month
1-subscription-started.json	testcases updated	last month
1-yp-subscripton.pcap	testcases updated	last month
2-push-update.json	testcases updated	last month
2-subscription-started.json	testcases updated	last month
2-yp-subscripton.pcap	testcases updated	last month
27-push-update.json	testcases updated	last month
27-subscription-started.json	testcases updated	last month
27-yp-subscripton.pcap	testcases updated	last month
readme.md	testcases updated	last month

readme.md

YANG-Push Notifications

subscription-started, push-update and push-change-update notifications in raw and json sorted by subscription id. pcap requires <https://github.com/network-analytics/udp-notif-dissector> Wireshark dissector.

YANG-Push UDP-Notif – DTLS and CBOR

- With Cisco IOS XR and 6Wind VSR we have now two implementations support [CBOR Name Identifiers](#)
- With Cisco IOS XR we have the first implementation support [DTLS 1.2 in UDP-Notif](#).

55	30.127022	10.215.132.90	10.212.242.71	DTLS	145 Client Hello
56	30.128282	10.215.132.90	10.212.242.71	DTLSv1.2	145 Client Hello
57	30.129707	10.215.132.90	10.212.242.71	DTLSv1.2	145 Client Hello


```

> Frame 56: Packet, 145 bytes on wire (1160 bits), 145 bytes captured (1160 bits)
> Linux cooked capture v1
> Internet Protocol Version 4, Src: 10.215.132.90, Dst: 10.212.242.71
> User Datagram Protocol, Src Port: 40308, Dst Port: 10101
v Datagram Transport Layer Security
  [Stream index: 55]
    v DTLSv1.2 Record Layer: Handshake Protocol: Client Hello
      Content Type: Handshake (22)
      Version: DTLS 1.0 (0xfef)
      Epoch: 0
      Sequence Number: 0
      Length: 88
    v Handshake Protocol: Client Hello
      Handshake Type: Client Hello (1)
      Length: 76
      Message Sequence: 0
      Fragment Offset: 0
      Fragment Length: 76
      Version: DTLS 1.2 (0xfef)
      Random: 2663bcc8809865a97d05adf677dbfaf74314780a050f7b7a9798d02a0bd280a8
      Session ID Length: 0
      Cookie Length: 0
      Cipher Suites Length: 4
      Cipher Suites (2 suites)
      Compression Methods Length: 1
      Compression Methods (1 method)
      Extensions Length: 30
      Extension: session_ticket (len=0)
      Extension: encrypt_then_mac (len=0)
      Extension: extended_master_secret (len=0)
      Extension: signature_algorithms (len=14)
        [JA4: dd21020400_a047181052cf_2e98e0d99bae]
        [JA4_r: dd21020400_009d_00ff_000d_0016_0017_0023_0804_0805_0806_0401_0501_0601]
        [JA3 Fullstring: 65277,157-255,35-22-23-13,,]
        [JA3: a613dbde97342f7533aa4b6edccc1673]
  
```

No.	Time	Source	Destination	Protocol	Length	Info
6	35.112763	127.0.0.1	127.0.0.1	UDPPUBCHANNEL	772	43006 → 10100 Len=728
7	35.120730	127.0.0.1	127.0.0.1	UDPPUBCHANNEL	709	54574 → 10100 Len=665


```

> Frame 7: Packet, 709 bytes on wire (5672 bits), 709 bytes captured (5672 bits)
> Linux cooked capture v1
> Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
> User Datagram Protocol, Src Port: 54574, Dst Port: 10100
> UDP-Notif Protocol, Publisher: 0, Msg ID: 71
v Concise Binary Object Representation
  Map: (indefinite length)
    101. .... = Major Type: Map (5)
    ...1 1111 = Size: Indefinite Length (31)
    v Text String: ietf-yp-notification:envelope
      011. .... = Major Type: Text String (3)
      ...1 1000 = Size: 1 byte (24)
      Length: 29
      Text String: ietf-yp-notification:envelope
    v Map: (indefinite length)
      101. .... = Major Type: Map (5)
      ...1 1111 = Size: Indefinite Length (31)
      v Text String: event-time
        011. .... = Major Type: Text String (3)
        ...0 1010 = Length: 10
        Text String: event-time
      v Text String: 2026-07-18T14:41:38.860Z
        011. .... = Major Type: Text String (3)
        ...1 1000 = Size: 1 byte (24)
        Length: 24
        Text String: 2026-07-18T14:41:38.860Z
      v Text String: hostname
        011. .... = Major Type: Text String (3)
        ...0 1000 = Length: 8
        Text String: hostname
      v Text String: ipf-zbl1327-r-daisy-90
        011. .... = Major Type: Text String (3)
        ...1 0110 = Length: 22
        Text String: ipf-zbl1327-r-daisy-90
      v Text String: sequence-number
        011. .... = Major Type: Text String (3)
        ...0 1111 = Length: 15
        Text String: sequence-number
      v Unsigned Integer: 71
  
```

YANG-Push Capabilities & YANG Library - RPC

- Netconf XML encoded reply for [RFC 9196](#) YANG-Push capabilities.
- Netconf XML encoded reply for [RFC 8525](#) with [draft-ietf-netconf-yang-library-augmentedby](#).


graf3net update
bf0be62 · 16 hours ago
 History

Name	Last commit message	Last commit date
 ..		
 get-systems-capability.xml	update	16 hours ago
 get-yang-library.xml	update	16 hours ago

YANG-Push

IETF 126 Publisher Implementation Status

	6WIND VSR	Huawei NE	Huawei MA	Cisco IOS XR	Arrcus Arcos	Open- Source	Ductus DMAP
RFC 8639 YANG-Push Subscription	✓	✓	✓	✓	✓		✓
RFC 8641 YANG-Push Notification	✓	✓	✓	✓	✓		✓
draft-ietf-netconf-udp-notif	✓	✓	✓	✓	✓	✓	
draft-ietf-netconf-udp-notif with DTLS				✓			
draft-ietf-netconf-distributed-notif	✓	✓	✓		na		
draft-ietf-netconf-notif-envelope	✓	✓	✓	✓	✓		P
draft-ietf-netconf-yang-notifications-versioning	✓	✓	✓	✓	✓		
RFC 8525 YANG Library	✓	✓	✓	✓	✓		✓
draft-ietf-netconf-yang-library-augmentation	✓	✓	✓	✓		✓	✓
RFC 9196 System and Notification Capabilities			✓	P	✓		✓
draft-netana-netconf-yp-transport-capabilities			✓	✓	✓		
RFC 9254 CBOR Named Identifiers	✓			✓			

Green marked describes new capability at IETF 125. "P" to partially implemented.

Operational IETF/IEEE YANG

IETF 126 Module Implementation Status

	6WIND VSR	Huawei NE	Huawei MA	Cisco IOS XR	Arrcus Arcos
ietf-interfaces.yang (inventory, state and statistic)	✓	✓	✓		
ietf-hardware.yang (inventory, state and statistic)	✓	✓	✓		
ietf-alarms.yang (state)	✓	✓	✓		
ieee802-dot1ab-lldp.yang (inventory and state)	✓	✓			
ieee802-dot1ax.yang (inventory, state and statistic)					
ietf-bfd-ip-sh.yang (state)					
ietf-bfd-ip-mh.yang (state)					
ietf-bfd-lag.yang (state)					
ietf-isis.yang (state)					

Green marked describes new capability at IETF 126. "P" to partially implemented.

YANG Message Broker

IETF 126 Implementation Status

	NetGauze™	Pmacct	Apache Kafka	Ciena Blueplanet	Cisco Crosswork
draft-ietf-nmop-message-broker-telemetry-message	✓	✓			
YANG-Push Receiver	✓	✓			
YANG Schema Retrieval	✓				
YANG Message Broker Producer	✓				
YANG Schema Registry	na	na	✓	na	na
YANG Message Broker Consumer	✓			✓	✓
Validate Telemetry Message against YANG Schema	✓			P	P
draft-ietf-netmod-yang-anydata-validation	✓				
YANG Schema Comparison	na	na	✓	na	na

Green marked describes new capability at IETF 126. "P" to partially implemented.

YANG Library Feature Comparison

- An extensive YANG feature comparison between the different YANG library open-source implementations.
- Focuses on YANG features needed to validate YANG instance data against YANG schema and performing YANG to YANG transformations.

Table 15. Global synthesis of support levels across evaluated feature families

Feature family	libyang	Yangkit	ODL Yang Tools
Schema loading and validation	Supported	Supported	Partially supported
Module inspection	Partially supported	Supported	Supported
JSON data validation	Supported	Supported	Partially supported
CBOR data validation	Not supported	Supported	Not supported
Anydata validation	Supported	Partially supported	Partially supported
YANG structures	Supported	Supported	Not supported
Anydata inside structures	Supported	Partially supported	Not applicable
Structures inside anydata	Supported	Supported	Not applicable
XPath/data extraction	Supported	Partially supported	Partially supported
Subtree validation	Supported	Supported	Supported
YANG-to-YANG transformation	Supported	Supported for the described add/update cases	Supported for the described add/update cases
YANG Library loading	Supported	Supported	Supported
Yang Push validation	Supported	Supported	Not applicable
Telemetry validation	Supported	Supported	Not applicable
Notification envelope validation	Supported	Supported	Not applicable
Schema comparison	Supported through specific yanglint branch	Implementable	Implementable

https://github.com/network-analytics/yang_tests/blob/master/report_v2.pdf

Next Steps from IETF 125 Hackathon...

- Re-do YANG Message Broker Consumer tests on updated testbed with Ciena Blue Planet UAA Workflow Engine
- Complete IETF YANG module subscription tests with 6Wind VSR
- Compare YANG feature coverage between [libyang](#), [yangkit](#) and [yangtools](#)
- Revert [import only modules for augmented-by](#) workaround
- [Preserve cached YANG-Push subscription state across receiver reload](#)
- [Obtain YANG-Push subscription state from publisher if missing](#)
- Support Netconf session queuing for schema retrieval
- Support libyang 5.4.9 in YANG-Push Receiver and Message Broker Consumer
- [Message Key and Topic Name proof of concept](#)



Next Steps for IETF 127 Hackathon...

- Support building [RFC 8525 YANG Library in Java Message Broker Consumer](#)
- Support [draft-ietf-netmod-yang-anydata-validation](#) and RFC 8525 YANG library in yangkit
- Support [YANG-Push Message Broker Topic Naming](#) with [dynamic Apache Kafka topic creation API](#)

Thanks to...

- Jiaxin PAN – Huawei
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- Jérémie Leska – 6Wind
- Per Anderson - Ductus
- Deepya Mandadi – Ciena
- Sivakumar Sundaravadivel – Ciena
- IJsbrand Wijnands – Arrcus
- Irfan Mohammad – Arrcus
- Benoit Claise - Everything OPS
- Paolo Lucente – Pmacct
- Vivekananda Boudia – INSA Lyon
- Alex Huang Feng – INSA Lyon/T-Systems
- Maxence Younsi – INSA Lyon/Swisscom
- Ashvanth Kumar Selvakumaran - Telus
- Leonardo Rodoni - Swisscom
- Ahmed Elhassany – Swisscom
- Thomas Graf – Swisscom

Want to know more then join us at...

- NMOP on **Monday 09:00 – 11:00** at the Grand Klimt Hall 2 for Message Broker, Message Key, BMP Telemetry Message document updates and hackathon related details.
- NETMOD on **Thursday 09:00 – 11:00** at the Grand Klimt Hall 1 for YANG Library feature comparison related details.